ICSE Class 10 Python
Lists and Tuples - Exam Ready Notes

1 Lists in Python

- A **list** is a **mutable** (changeable), **ordered** sequence of elements.
- Created using **square brackets** `[]`.
- Can store elements of **different data types** (heterogeneous).
- Supports **indexing** (starts at 0) and **negative indexing**.

1.1 Examples

```
# Empty list
l1 = []

# List with different types
l2 = [10, 20.5, "Hello", True, [1,2,3]]

# Accessing elements
print(l2[0])      # 10
print(l2[-1])     # [1, 2, 3]
print(l2[2:4])    # ['Hello', True]  # Slicing

```

1.2 List Operations

- Concatenation: `list1 + list2`
- Repetition: `list * n`
- Membership: `element in list`
- Length: `len(list)`

1.3 Important List Methods

Method	Description	Example
`append(x)`	Adds element at the end	`lst.append(50)`
`insert(i, x)`	Inserts at index i	`lst.insert(1, 25)`
`extend(lst2)`	Adds all elements of lst2	`lst.extend([60,70])`
`remove(x)`	Removes first occurrence of x	`lst.remove(20)`
`pop(i)`	Removes and returns element at i	`lst.pop()`
`clear()`	Removes all elements	`lst.clear()`
`index(x)`	Returns index of first occurrence	`lst.index(30)`
`count(x)`	Counts occurrences of x	`lst.count(10)`
`sort()`	Sorts in ascending order	`lst.sort()`
`reverse()`	Reverses the list	`lst.reverse()`

1.4 Nested Lists

```
nested = [[1,2,3], [4,5,6], [7,8,9]]
print(nested[1][2])  # 6

```

2 Tuples in Python

- A **tuple** is an **immutable** (cannot be changed), **ordered** sequence.
- Created using **parentheses** `()` or comma separated values.
- Faster than lists; used for fixed data.

2.1 Examples

```
t1 = (10, 20, 30)
t2 = 1, 2, 3, 4          # Parentheses optional
t3 = ("apple",)          # Single element needs comma
empty = ()               # Empty tuple

```

2.2 Tuple Assignment / Packing-Unpacking

```
# Packing
t = 10, 20, 30

# Unpacking
a, b, c = t
print(a, b, c)  # 10 20 30

# Swap variables
x, y = y, x

```

3 Key Differences: List vs Tuple

Feature	List	Tuple
Mutability	Mutable	Immutable
Syntax	`[]`	`()`
Speed	Slower	Faster
Use case	Changing data	Fixed data
Memory	More	Less

4 Common Functions

- `len()` → length
- `max()`, `min()` → on numeric sequences
- `sum()` → for numbers
- `sorted(seq)` → returns new sorted list
- `list(tuple)` / `tuple(list)` → type conversion